



Open Cosmos VNIR-250 L1D: Quality Processes Review

Author(s): *Samantha Malone*
NPL

Reviewer(s): *Sam Hunt*
NPL

Contents

1	Introduction	2
1.1	Scope	2
1.2	Changes Log.....	2
1.3	Applicable Documents.....	2
1.4	Reference Documents	2
1.5	Acronyms & Abbreviations.....	2
2	Review Summary – Product Maturity Matrix	4
3	Product Information	5
4	Metrology.....	9
5	Product Generation	11

1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for OpenCosmos VNIR-250 L1D as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
1.0	03/03/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1]	Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. Available Online .
[RD-2]	OpenApp VNIR-250 L1D Data Specification. See: https://app.open-cosmos.com/help/data-catalog/products/vnir-250
[RD-3]	eoPortal, Satellite Missions Catalogue. See: https://www.eoportal.org/satellite-missions/mantis
[RD-4]	Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016); https://doi.org/10.1038/sdata.2016.18
[RD-5]	https://cogeo.org/

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration

NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- *Not Assessable* means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for OpenCosmos VNIR-250 L1D is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation		Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for OpenCosmos VNIR-250 L1D

3 Product Information

Product Details			
Grade: Basic			
Justification	Some information supplied in the OpenApp VNIR-250 Data Specification [RD-2] and at eoPortal [RD-3], however many important pieces of information missing, including stated measured quantity and units, measurement quality, spatial coverage and temporal resolution.		
Product Name	OpenCosmos VNIR-250 L1D		
Sensor Name	VNIR-250		
Sensor Type	Optical – Multispectral (4 bands) in the visible and NIR range.		
Mission Type	1 satellite (MANTIS)		
Mission Orbit	Sun-synchronous, with an altitude of 500 km, and a Local Time of Descending Node at 10:30 hours.		
Product Version Number	Not stated.		
Product ID	The file names for VNIR-250 data products follow a standard convention as defined on the OpenApp Data Specification:		
	Field	Description	Example
	Mission	Mission name	MANTIS
	Level	Processing level	L1C, L1D
	Session + frames	4-digit session + 4-digit start frame + 4-digit end frame	000003260363 (session 0000, frames 326–363)
	Start / End	Acquisition start/end (UTC, YYYYMMDDThhmmss)	20250504T181431 / 20250504T181432
	e.g. MANTIS_L1C_000X03260363_20250504T181431_20250504T181432		
Processing level of product	MANTIS L1D data products are defined as ortho-ready high-resolution imagery.		
Measured Quantity Name	TOA data – not specifically stated.		
Measured Quantity Units	Not stated.		
Stated Measurement Quality	Radiometric Quality: Not stated. Geometric Quality: Not stated.		
Spatial Resolution	High-resolution RGB Ground Sampling Distance (GSD): 2.2 m at 500 km NIR Ground Sampling Distance (GSD): 3.0 m at 500 km		
Spatial Coverage	Orbital period: Not stated. Orbit inclination: Not stated.		
Temporal Resolution	Revisit capacity: Not stated.		

Temporal Coverage	<i>Mission lifetime: ~2 years. Launched: 11th November 2023 De-orbited: 22nd October 2025</i>
Point of Contact	<i>sales@open-cosmos.com</i>
Product locator (DOI/URL)	<i>Products available on request via https://www.open-cosmos.com/data-and-services#Contact-Form</i>
Conditions for access and use	<i>Not stated.</i>
Limitations on public access	<i>Not publicly available.</i>
Product Abstract	<i>From [RD-2]:</i> <i>The VNIR-250 product offers high-resolution imagery in the Visible and Near-Infrared spectrum.</i> <i>The optical payload is an iSIM-90 integrated Standard Imager developed by SATLANTIS and is a new generation of high-resolution multi-spectral optical imager for earth observation developed for 12U or larger CubeSats.</i> <i>The multispectral payload captures 4 channels in the visible and near infrared spectral range. These channels align with the corresponding channels in Sentinel 2 imagery.</i>

Availability & Accessibility	
Grade: Basic	
Justification	<i>The product information provided, such as Product Locator and Conditions for access and use, meets only some of the FAIR principles.</i> <i>Findable, Accessible, Interoperable and Reusable (FAIR) Principles – see [RD-4].</i>
Compliant with FAIR principles	No
Data Management Plan	Description of data management plan is not available.
Availability Status	Products available on request via https://www.open-cosmos.com/data-and-services#Contact-Form

Product Format, Flags & Metadata				
Grade: Good				
Justification	COG (Cloud Optimized Geotiff) [RD-5], a documented standard file format, is used. The products also include a good set of documented metadata with a self-defined standard naming convention, but data flag information is not made available. Product is not CEOS ARD compliant.			
Product File Format	Products are available in a well-defined file format structure defined at [RD-2]:			
	Title	Key	File Format	Description
	Band data	B, G, R, NIR etc	COG	Cloud optimised raster data containing band spectral data
	Rational Polynomial Coefficients (RPCs)	"RPC"	"_rpc.txt", CE NITF v2.1	Text file containing RPCs for RGB and NIR images following the National Imagery Transmission Format (NITF) standard Version 2.1
	Thumbnail (TCI)	"thumbnail"	webp	A low resolution true colour composite image for quick visualisation of imagery data, typically used as a preview in the browser
Metadata Conventions	STAC			
Analysis Ready Data?	No			

User Documentation		
Grade: Basic		
Justification	There is no specific user guide, but a Data Specification page is available with high-level information (see [RD-2]). There is no ATBD made publicly available.	
<i>Document</i>	<i>Reference</i>	<i>QA4ECV Compliant</i>
Product User Guide	N/A	No
ATBD	<i>Documentation not made available to users</i>	N/A

4 Metrology

Sensor Calibration & Characterisation	
Grade: Not Assessable	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly and therefore it is not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	

Geometric Calibration & Characterisation	
Grade: Not Assessable	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly and therefore it is not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	

Metrological Traceability Documentation	
Grade: Not Assessable	
Justification	<i>There are no metrology documentation or traceability chain / uncertainty tree diagrams publicly available.</i>
References	

Uncertainty Characterisation	
Grade: Not Assessable	
Justification	<i>Uncertainty information is not publicly available nor is any detailed information regarding the methods used to characterise the uncertainties publicly available.</i>
References	

Ancillary Data	
Grade: Not Assessable	

Justification	<i>There is information on the format of the ancillary data but no description of what specific ancillary data is included with the product and therefore there is neither any detailed information regarding uncertainty characterisation for any of the ancillary data.</i>
References	

5 Product Generation

Calibration Algorithm	
Grade: Not Assessable	
Justification	<i>Documentation states that to obtain the L1D product, the data is radiometrically corrected to Top-of-Atmosphere (TOA), however, any details on the methods used to evaluate these corrections is not available publicly and it is therefore not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	• RD-2

Geometric Processing	
Grade: Not Assessable	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly.</i> <i>There is some very high-level information at [RD-2] which outlines the main geometric processes that have been carried out to obtain L1D:</i> • <i>Geometric correction to an accuracy of ~100m</i> • <i>Pan sharpening</i> • <i>Ortho-ready, metadata files containing Rational Polynomial Coefficients (RPCs)</i> <i>However, any details on the methods used to evaluate these corrections is not available publicly and it is therefore not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	• RD-2